Worksheet 8 Solution

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1. Acme Corporation's helpdesk gets 4 calls per day on average. They think the number of calls follows a Poisson distribution.

   a. What is the probability that they get 3 calls or less on a given day?

   b. What is the probability that they get no calls on given day?

   c. What is the probability that they get exactly 3 calls?

   d. What is the expected number of calls in a week?

   e. What is the standard deviation for calls in a day?

   <u>Answer:</u>

   (a) Using table: $P[x \leqslant 3] = 0.4335$.

   (b) Using table:

   $$P[x = 0] = P[x \leqslant 0] = 0.0183$$

   Direct calculation:

   $$P[x = 0] = \frac{e^{-4}4^0}{0!} = 0.01831564$$

   (c) Using table: $P[x = 3] = P[x \leqslant 3] - P[x \leqslant 2] = 0.4335 - 0.2381 = 0.1954$.

   Direct calculation: $P[x = 3] = \frac{e^{-4}4^3}{3!} = 0.1953668$

   (d) Let $Y = \#$ calls per week on average. Then, $Y = 7X$. $E(Y) = E(7X) = 7E(X) = 7 \times 4 = 28$.

   (e) $V(X) = \mu = 4 \rightarrow SD(x) = \sqrt{4} = 2$.

2. The number of flaws in an optic fiber cable follows a Poisson Distribution. The average number of flaws in 50 m is 1.5. Let X = average number of flaws in 50 m.

   a. What is the probability of exactly 2 flaws in 100 m ?

   b. What is the probability of 3 flaws or less in 150 m ?

   <u>Answer:</u>

   (a) Let Y = average number of flaws in 100 m. $Y = 2X$. $E(Y) = 2E(X) = 3$. Using table:

$$P(Y = 2) = P(Y \leqslant 2) - P(Y \leqslant 1) = 0.4232 - 0.1991 = 0.2241$$

   Or, using direct calculation:

$$P(Y = 2) = \frac{e^{-3}3^2}{2!} = 0.2240418$$

   (b) Let Z = average number of flaws in 150 m. $Z = 3X$. $E(Z) = 3E(X) = 4.5$. Using table:

$$P(Y \leqslant 3) = 0.3423$$

3. X is a Uniformly distributed random variable that has maximum and minimum values of 10 and 50.

   a. What is the density function and its graph?

   b. Calculate the mean of X.

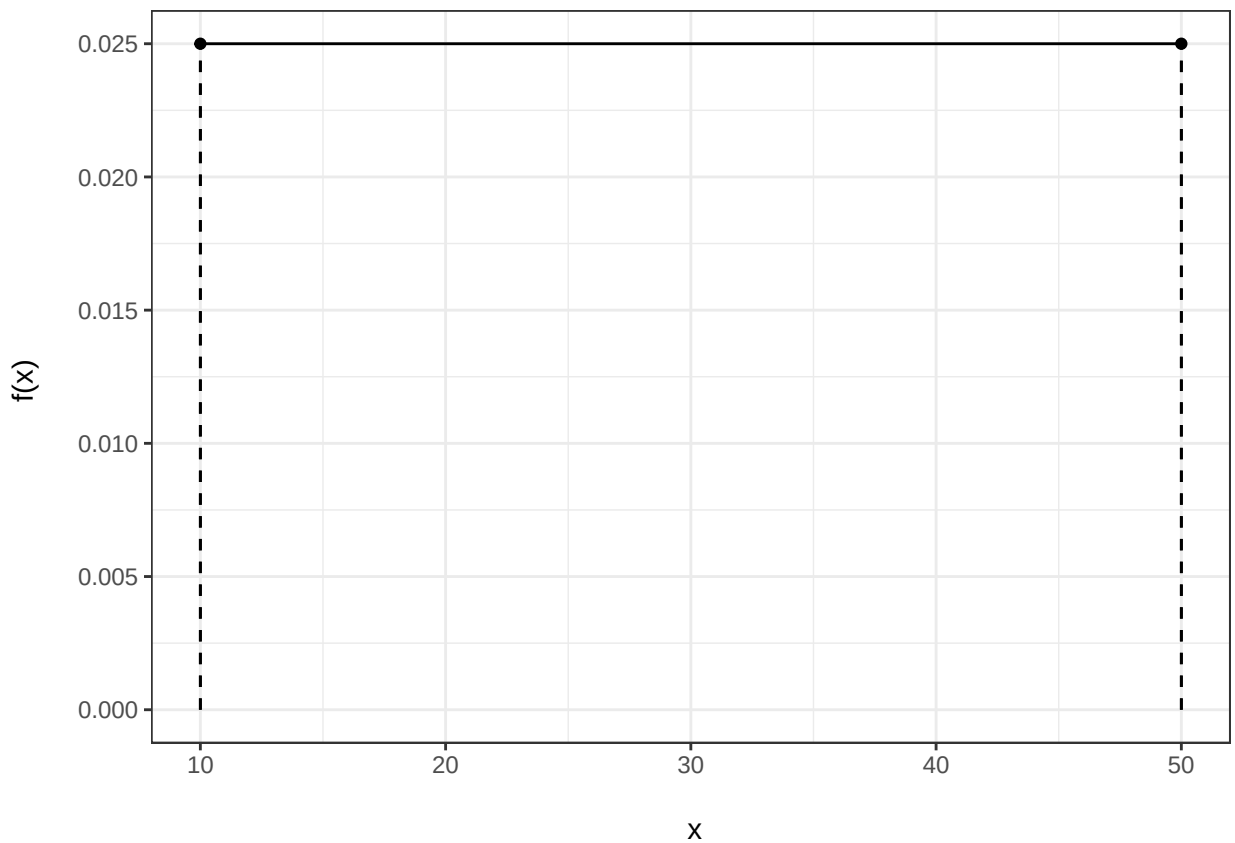
   c. Calculate $P[10 < X < 20]$ and mark the corresponding area on the graph from (a).

   d. Calculate $P[25 < X < 45]$ and mark the corresponding area on the graph from (a).

   e. Find $P[X = 22.5]$.

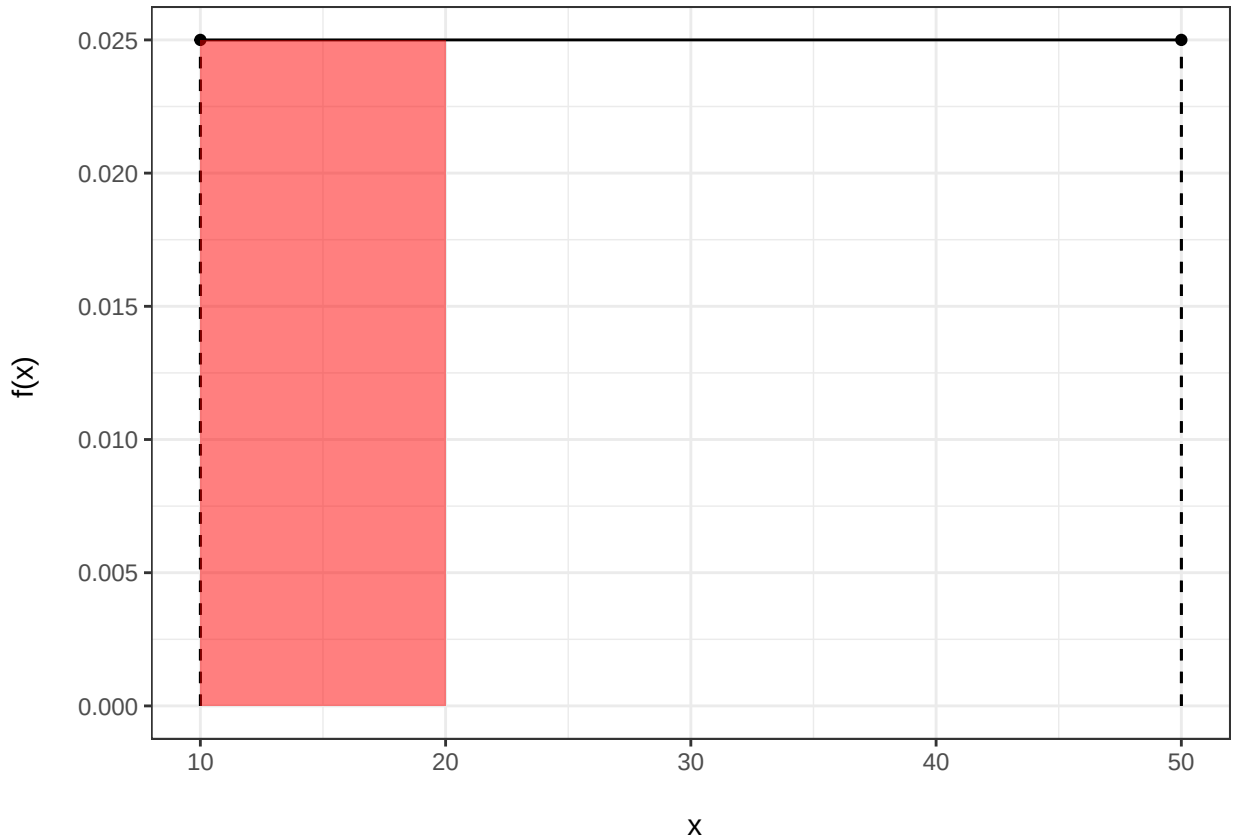
<u>Answer:</u>

(a) $f(x) = \frac{1}{b-a} = \frac{1}{50-10} = \frac{1}{40}$.

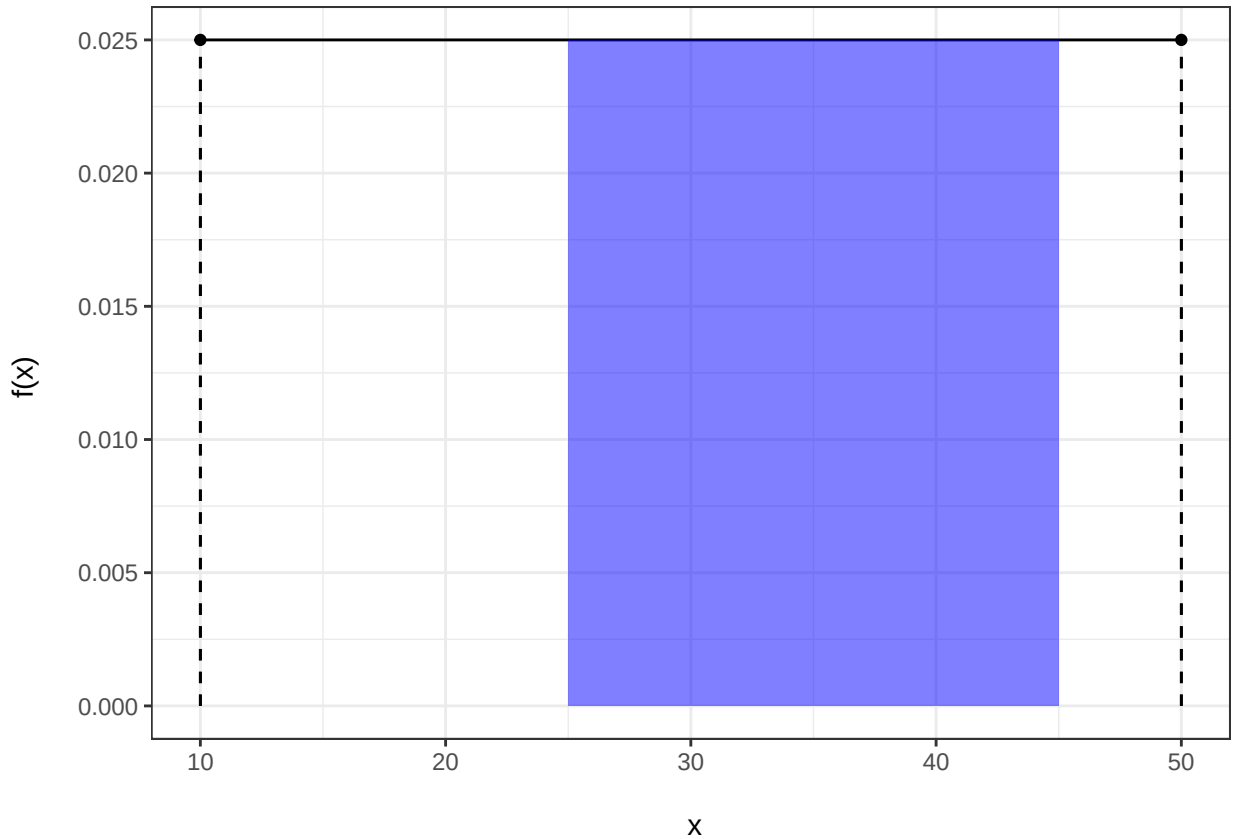


(b) $\mu = \frac{a+b}{2} = \frac{10+50}{2} = 30$.

(c) Blue area= $(20 - 10) \times \frac{1}{40} = \frac{1}{4} = 0.25$.



(d) Red area= $(45 - 25) \times \frac{1}{40} = \frac{1}{2} = 0.5$.



(e) This is a continuous RV. Hence, P[X=22.5]=0$.

1. Calculate the following Probabilities from using a standard Normal distribution, $Z \sim$ N(0,1).

   a. $P[Z < 0]$.

      A: Using table, $P[Z < 0] = 0.5$.

   b. $P[Z < -1.5]$.

      A: Using table, $P[Z < -1.5] = 0.0668$.

   c. $P[Z > 1]$.

      A: $P[Z > 1] \overset{\text{Compliment R}}{=} 1 - P[Z \leq 1] = 1 - 0.8413 = 0.1587$.

   d. $P[Z > -1.44]$.

      A: $P[Z > -1.44] \overset{\text{Compliment R}}{=} 1 - P[Z \leq -1.44] = 1 - 0.0749 = 0.9251$.

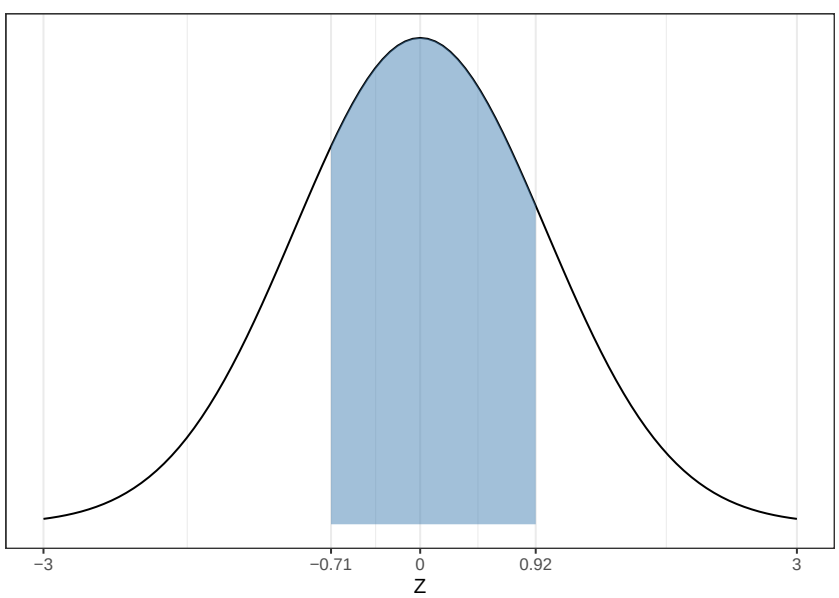
   e. $P[Z > 3.09]$.

      A: $P[Z > 3.09] \overset{\text{Compliment R}}{=} 1 - P[Z \leq 3.09] = 1 - 0.9990 = 0.001$.

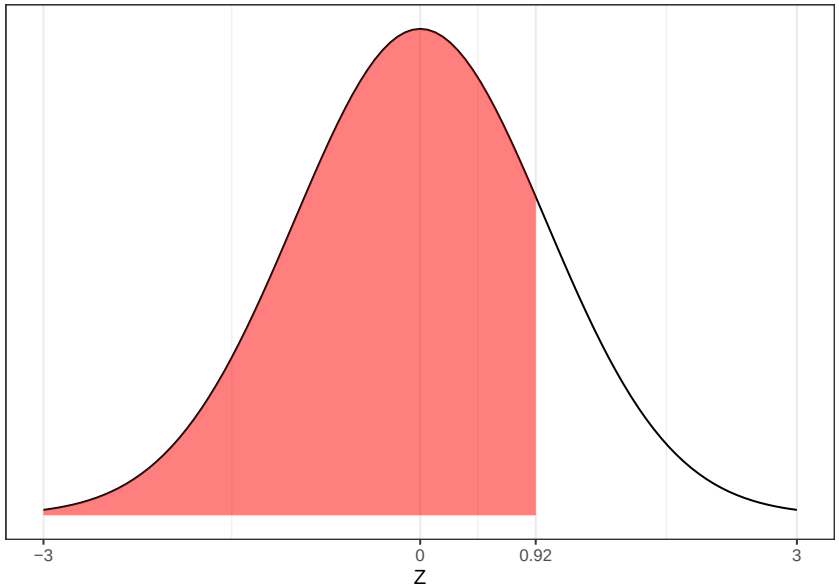
   f. $P[-0.71 < Z < 0.92]$.

      A: $P[-0.71 < Z < 0.92] = P[Z < 0.92] - P[Z < -0.71] = 0.8212 - 0.2389 = 0.5823$.
      Why? See below:

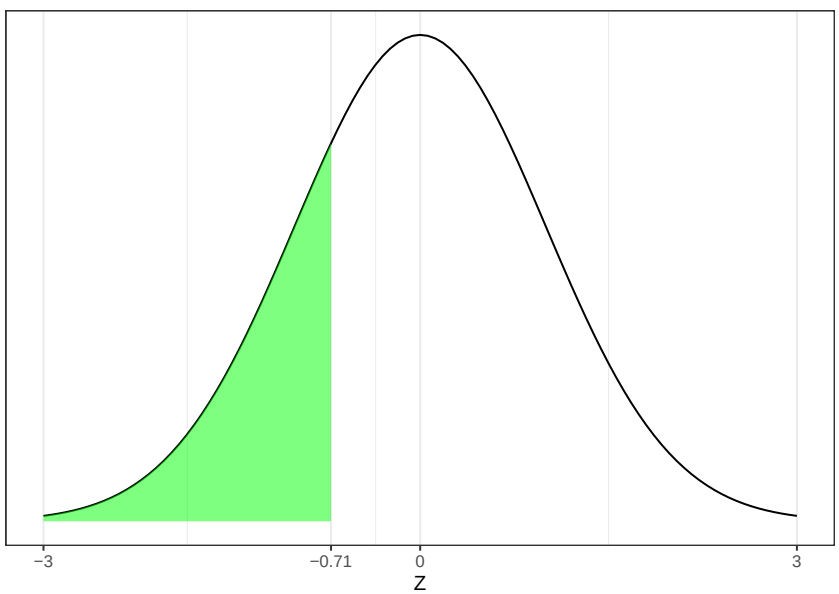
      This is what we want:



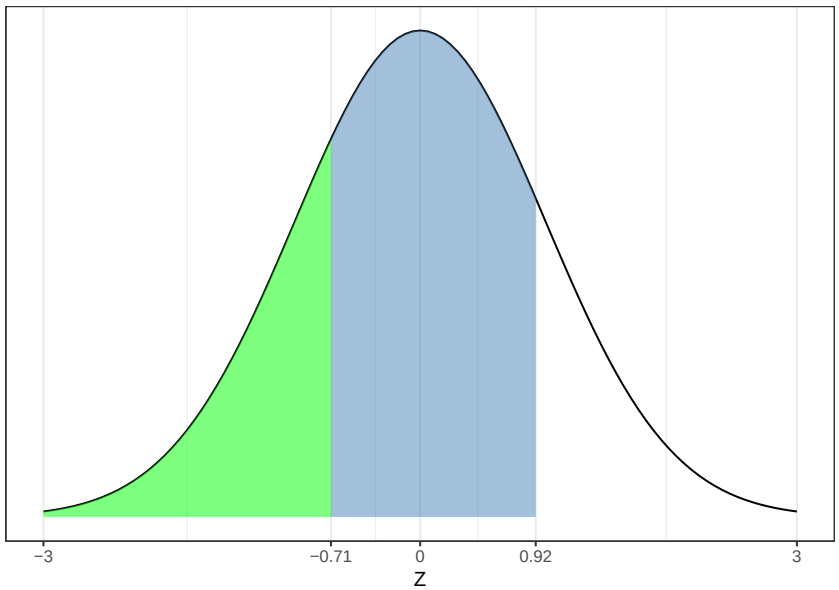
We note that P[Z<0.92] would be the red area on this plot:



We also note that P[Z<-0.71] would be the green area on this plot:



If I get rid of the green area in the red area, what I am left with would be the blue area.



g. $P[-1.5 < Z < -1.44]$.

A: $P[-1.5 < Z < -1.44] = P[Z < -1.44] - P[Z < -1.5] = 0.0749 - 0.0668 = 0.0081$.

h. $P[Z < -0.71 \text{ or } Z > 0.92]$.

A: $P[Z < -0.71 \cup Z > 0.92] = P[Z < -0.71] + P[Z > 0.92] - P[Z < -0.71 \cap Z > 0.92]$. There is no common Z in the intersection, i.e $P[Z < -0.71 \cap Z > 0.92] = 0$. Hence:

$$\begin{aligned} P[Z < -0.71 \cup Z > 0.92] &= P[Z < -0.71] + P[Z > 0.92] \\ &= P[Z < -0.71] + 1 - P[Z \leq 0.92] \\ &= 0.2389 + 1 - 0.8212 \\ &= 0.4177 \end{aligned}$$